5.06 The fetch–execute (F–E) cycle

The full name for this is the fetch, decode and execute cycle. This is illustrated by the flowchart in Figure 5.03.

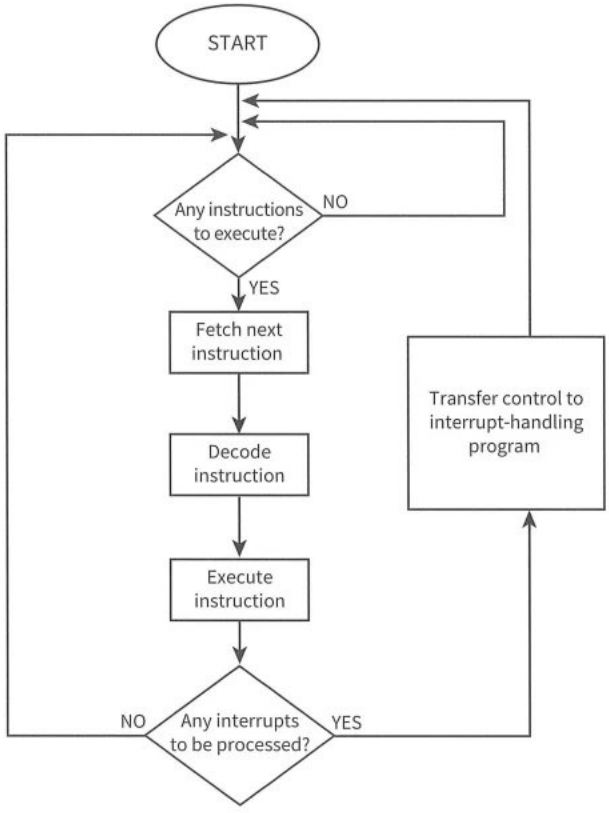


Figure 5.03 Flowchart for the fetch, decode and execute cycle

If we assume that a program is already running, then the program counter will already hold the address of the next instruction. In the fetch stage, the following steps will now happen.

1. This address in the program counter is transferred within the CPU to the MAR.
2. During the next clock cycle two things happen simultaneously:
   - the instruction held in the address pointed to by the MAR is fetched into the MDR
   - the address stored in the program counter is incremented.
3. The instruction stored in the MDR is transferred within the CPU to the CIR.

   There are two points to note here.
   - The clock cycle is the one controlled by the system clock which will have settings that allow one data transfer from memory to take place in the time defined for one cycle.
   - In the final step the program counter is incremented by 1. However, the instruction just loaded might be a jump instruction. In this case, the program counter contents will have to be updated in accordance with the jump condition. This can only happen after the instruction has been decoded.

In the decode stage, the instruction stored in the CIR is received as input by the circuitry within the control unit. Depending on the type of instruction, the control unit will send signals to the appropriate components so that the execute stage can begin. At this stage, the ALU will be activated if the instruction requires arithmetic or logic processing.

The description of the execute stage is given in Chapter 6, where a simple instruction set is introduced and discussed.

5.07 Register transfer notation

Operations involving registers can be described by register transfer notation. A simple example of this is a representation of the fetch stage of the fetch–execute cycle:

```
MAR ← [PC]

PC ← [PC] + 1; MDR ← [[MAR]]

CIR ← [MDR]
```

In register transfer notation the basic format for an individual data transfer is similar to that for variable assignment. The first item is the destination for the data. Here the appropriate abbreviation is used to identify the particular register. To the right of the arrow showing the transmission of data is the definition of this data. In this definition, the square brackets around a register abbreviation show that the content of the register is being moved. This movement might also include an arithmetical operation. When two data operations are placed on the same line separated by a semi-colon, this means that the two transfers take place simultaneously. The double pair of brackets around MAR on the second line needs careful interpretation. The content of the MAR is an address; it is the content of that address which is being transferred to the MDR.

5.08 Interrupt handling

There are many different reasons for an interrupt to be generated. Some examples are:

- a fatal error in a program
- a hardware fault
- a need for I/O processing to begin
- user interaction
- a timer signal.

Discussion Point:

Carry out an investigation into the different causes of an interrupt.

Interrupts are handled by a number of different mechanisms, but there are some clear overriding principles. Each different interrupt needs to be handled appropriately. Different interrupts might have different priorities. Therefore, the processor must have a means of identifying the type of interrupt. One way is to have an interrupt register in the CPU that works like the status register, with each individual bit operating as a flag for a specific type of interrupt.

As the flowchart in Figure 5.03 shows, the existence of an interrupt is only detected at the end of a fetch–execute cycle. This allows the current program to be interrupted and left in a defined state which can be returned to later. An interrupt is handled by the following steps.

- The contents of the program counter and any other registers are stored somewhere safe in memory.

- The appropriate interrupt handler or Interrupt Service Routine (ISR) program is initiated by loading its start address into the program counter.
- When the ISR program has been executed there is an immediate check to see if further interrupts need handling.
- Further interrupts are dealt with by repeated execution of the ISR program.
- If there are no further interrupts, the safely stored contents of the registers are restored to the CPU and the originally running program is resumed.

Reflection Point:

Have you worked out a method to remember all of the names and abbreviations for the special purpose registers?

Summary

- The von Neumann architecture for a computer system is based on the stored program concept.
- The CPU contains a control unit, an arithmetic and logic unit and registers.
- Registers can be special purpose or general purpose.
- The status register has individual bits acting as condition flags.
- The system bus contains the data, address and control buses.
- A universal serial bus (USB) port can be used to attach peripheral devices.
- Instructions are handled by the fetch–execute cycle.
- Register transfer notation is used to describe data transfers.
- If an interrupt is detected, control passes to an interrupt-handling routine.

Exam-style Questions

1 **a** A processor has just one general-purpose register. Give the name of this register. [1]

b The memory address register (MAR) is a special-purpose register. State:

i its function

ii the type of data stored in it

iii the register that supplies these data at the start of the fetch stage of the fetch–execute cycle. [3]

c The current instruction register (CIR) is another special-purpose register. State:

i its function

ii the type of data stored in it

iii the register that supplies this data at the end of the fetch stage of the fetch–execute cycle. [3]

d Explain **three** differences between the memory address register and the memory data register. [5]

2 The system bus comprises of three individual buses: the data bus, the address bus and the control bus.

a For each bus give a brief explanation of its use. [6]

b Each bus has a defined bus width.

i State what determines the width of a bus. [1]

ii Explain which bus will have the least width. [2]

iii Explain the effect of changing the address bus from a 32-bit bus to a 64-bit bus. [3]

3 The fetch stage of the fetch–decode–execute cycle can be represented by the following statements using register transfer notation:

```
MAR ← [PC]
PC ← [PC] + 1; MDR [[MAR]]
CIR ← [MDR]
```

a Explain the meaning of each statement. The explanation must include definitions of the following items: MAR, PC, [ ], , MDR, [[ ]], CIR. [10]

b Explain the use of the address bus and the data bus for two of the statements. [4]

4 **a** Name and describe **three** buses used in the von Neumann model. [6]

b The sequence of operations shows, in register transfer notation, the fetch stage of the fetch-execute cycle.

```
1  MAR ← [PC]
2  PC  ← [PC] + 1
3  MDR ← [[MAR]]
4  CIR ← [MDR]
```

- `[register]` denotes contents of the specified register or memory location
- step 1 above is read as "the contents of the Program Counter are copied to the Memory Address Register"

i Describe what is happening at step 2. [1]

ii Describe what is happening at step 3. [1]

iii Describe what is happening at step 4. [1]

c Describe what happens to the registers when the following instruction is executed:

```
LDD 35
```

[2]

d **i** Explain what is meant by an interrupt. [2]

ii Explain the actions of the processor when an interrupt is detected. [4]

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5 **a** Describe how special purpose registers are used in the fetch stage of the fetch-execute cycle. [4]

b Use the statements A, B, C and D to complete the description of how the fetch-execute cycle handles an interrupt.

A	the address of the Interrupt Service Routine (ISR) is loaded to the Program Counter (PC).
B	the processor checks if there is an interrupt.
C	when the ISR completes, the processor restores the register contents.
D	the register contents are saved.

At the end of the cycle for the current instruction .....

If the interrupt flag is set, ..., ... and .....

The interrupted program continues its execution. [4]

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